

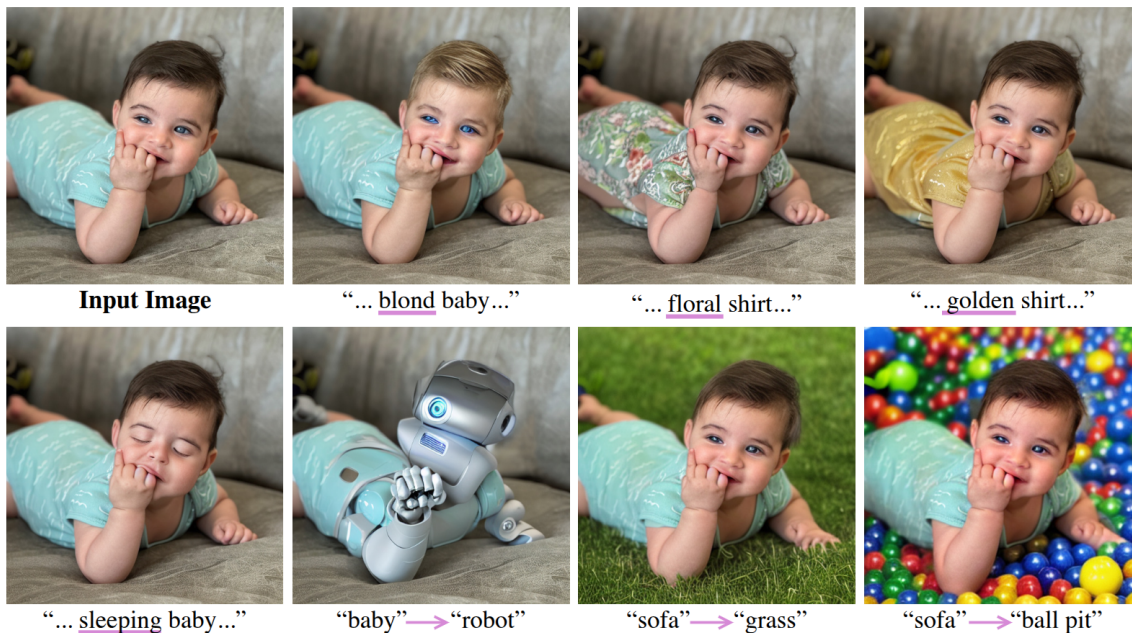
Image Editing with Diffusion Models

Exercise 2.3

Advanced Deep Learning in Computer Vision

March 2026

Input caption: “A baby wearing a blue shirt lying on the sofa.”



In this exercise, you will implement and experiment with two popular approaches for image editing using diffusion models: DDIM and Null-Text Inversion. You will start by implementing cross-attention visualization to understand how different tokens in the prompt influence different regions of the image. Then, you will implement the sampling and inversion processes of DDIM, experiment with different hyperparameters to understand their effects on the reconstructed images, and implement Null-Text Inversion to compare its reconstruction capabilities with those of DDIM inversion. Finally, you will apply these techniques to edit a picture.

Your tasks are as follows:

1. Complete the implementation of cross-attention visualization in `vis_cross_attn.py` to understand how different tokens in the prompt influence different regions of the generated image.
2. Complete the implementation of DDIM sampling in `ddim_sampling.py`.
3. Complete the implementation of DDIM inversion in `ddim_inversion.py`.
4. Complete the implementation of Null-Text Inversion in `null_text_inversion.py`.

5. Compare DDIM inversion and NTI by evaluating the reconstruction quality of the original image against the one obtained from DDIM inversion. Perform a quantitative evaluation using a test set with more than 10 images. Report average PSNR, SSIM, LPIPS, etc. between reconstructions and the original images.
6. Now, take a picture of your group and use `ddim_inversion_simple_editing.py` to make a fun edit, e.g., add sunglasses. Report your prompts and hyperparameters. Also, try different hyperparameters (number of timesteps, initial timestep, guidance scale). Report your findings and the effect of each hyperparameter on the reconstructed images.
7. **OPTIONAL:** Cannot we just add noise to the input image and denoise with the new prompt? Why bother inverting, why not just use an image-to-image pipeline? Complete the implementation in `image_to_image_simple_editing.py`. Compare this approach against the DDIM inversion+editing approach. Vary `start_step` and explain how the results differ.
8. **OPTIONAL:** Implement Prompt-to-Prompt method on top of Null-Text Inversion. Implement the cross-attention control technique by completing `cross_attention_editing.py`. Compare this approach against the previous editing methods you tried.
9. **OPTIONAL:** Measure editing accuracy quantitatively. Compare editing performances for different approaches you implemented using replacement-type editing operations on your test set. Hint: you can use vision-language models to evaluate the editing accuracy.
10. Your data, process, performance evaluation, and results should be documented and discussed in a PDF (up to 4 pages) to be uploaded on DTU Learn together with exercises 2.1 and 2.2. More details are in the given template.